

Theoretical Foundation

1. What is a Statistical Distribution?

A statistical distribution is a mathematical function that describes how the values of a dataset are spread or distributed. It shows the frequency or probability of different values occurring in the data.

Why is it important?

Understand Patterns

Understands the pattern of data.

Choose Tests

Helps choose appropriate statistical tests.

Predict Observations

Predicts future observations.

Support Decisions

Supports machine learning and business decision-making.

Example

Suppose the transaction amounts are:

₹200 ₹500 ₹750 ₹900 ₹1000 ₹1200 ₹5000

Most values are below ₹1500, while a few are very high. This indicates a **right-skewed distribution**.

2. What is a Q-Q Plot and Why is it Used?

A Q-Q (Quantile-Quantile) Plot is a graphical method used to compare the distribution of a dataset with a theoretical distribution, usually the Normal Distribution.

Purpose

- Check whether data follows a normal distribution.
- Detect skewness and outliers.
- Validate assumptions before statistical tests such as Regression, ANOVA, and t-tests.

Interpretation

If points lie close to the diagonal line:

✓ Data is approximately normally distributed.

If points deviate from the line:

Data is not normally distributed.

3. Difference Between Discrete and Continuous Distributions

Feature	Discrete Distribution	Continuous Distribution
Values	Countable integers	Infinite decimal values
Nature	Counting	Measuring
Function Used	PMF (Probability Mass Function)	PDF (Probability Density Function)
Examples	Number of transactions	Transaction amount

Discrete

- Number of successful transactions
- Number of customers

Continuous

- Transaction amount
- Time taken for payment

4. What is a Bernoulli Distribution?

A Bernoulli Distribution represents an experiment with only two possible outcomes.

Success (1)

Failure (0)

Characteristics

Only one trial

Two outcomes

Probability of success = p

Probability of failure = $1 - p$

Example

Transaction Status

Status	Value
Success	1
Fail	0

If 95% of transactions are successful:

$P(\text{Success}) = 0.95$

$P(\text{Failure}) = 0.05$

The transaction status column follows a Bernoulli Distribution.

5. What is a Binomial Distribution?

A Binomial Distribution models the number of successes obtained from a fixed number of independent Bernoulli trials.

Characteristics

- Fixed number of trials (n)
- Two outcomes per trial
- Constant probability of success
- Independent trials

Formula

$X \sim \text{Binomial}(n, p)$

where

n = number of trials

p = probability of success

Example

Suppose a customer makes 10 transactions in one week.

Question: How many transactions are successful?

Possible outcomes:

0 1 2 ... 10

The weekly transaction count can be modeled using a Binomial Distribution.

6. Explain Log-Normal Distribution

A variable follows a Log-Normal Distribution if the logarithm of that variable follows a Normal Distribution.

Mathematically,

If **$\log(X)$** is normally distributed, then **X** is log-normally distributed.

Characteristics

- Values are always positive.
- Right-skewed.
- Many small values and few very large values.

Applications

- Transaction amounts
- Income
- Stock prices
- House prices
- Insurance claims

Example

Most customers spend:

₹500 ₹700 ₹900 ₹1100

A few customers spend:

₹20,000 ₹40,000 ₹75,000

This pattern often follows a Log-Normal Distribution.

7. Explain Power Law Distribution

A Power Law Distribution describes situations where many observations are small, while a few observations are extremely large.

Characteristics

- Heavy-tailed distribution
- Very few extremely large values
- Scale-free behavior

Formula

$$P(x) = Cx^{-\alpha}$$

where

C = constant

α = exponent

Applications

Wealth distribution

Social media followers

Website traffic

Cryptocurrency holdings

Trading volume

Example

Customer transaction amounts:

₹100 ₹150 ₹250 ₹300 ₹500 ₹1000 ₹200000

Most customers spend small amounts, while a few spend exceptionally large amounts.

8. What is a Box-Cox Transform?

The Box-Cox Transform is a mathematical transformation used to make data more normally distributed and stabilize variance.

Why is it used?

- Reduce skewness
- Stabilize variance
- Improve regression performance
- Improve statistical testing accuracy

Advantages

- Makes data closer to Normal Distribution.
- Improves model assumptions.
- Reduces heteroscedasticity.



Limitation: It can only be applied to positive values.

9. Explain Poisson Distribution with an Example

A Poisson Distribution models the number of times an event occurs within a fixed interval of time or space.

Characteristics

- Counts occurrences.
- Events occur independently.
- Average rate remains constant.

Formula

$X \sim \text{Poisson}(\lambda)$

where

λ = average number of events

Example

Suppose an online store receives an average of 80 transactions per day.

Possible daily transactions:

75 78 81 84 90

The number of transactions per day can be modeled using a Poisson Distribution.

Other examples:

Calls received per hour

Website visitors per minute

Emails received per day

10. What is Z-score Probability?

A Z-score measures how far a data value is from the mean in terms of standard deviations.

Formula

$$Z = (X - \mu) / \sigma$$

where

X = observed value

μ = mean

σ = standard deviation

Interpretation

Z-score	Meaning
0	Equal to the mean
Positive	Above the mean
Negative	Below the mean
Large positive	Unusually high value
Large negative	Unusually low value

Example

Mean transaction amount = ₹3000

Standard deviation = ₹1000

Transaction amount = ₹5000

$$Z = (5000 - 3000) / 1000 = 2$$

This means the transaction amount is **2 standard deviations above the mean**.